

Invariant-Constrained Transition Framework (ICTF)

A Formal Measurement Framework for Consequential Agent Systems · Working Preprint Summary · Oct. 2026 · Not peer reviewed

Governance defines the admissible transition space. Optimization chooses within it.

Core Contribution

ICTF is a measurement and runtime-governance framework for agentic systems that participate in consequential state changes (tool calls, transactions, record updates, multi-agent coordination). It does not claim a universal alignment theory. Its purpose is to make distinct failure modes separately measurable rather than collapsing them into a single safety or alignment score. The framework separates five objects often conflated: (1) world state vs. admitted representation, (2) proposal representation vs. permit-bound executable action, (3) operational gate judgment vs. independent evaluation criterion, (4) local predicates vs. history-dependent trajectory invariants, and (5) model-relative compliance vs. the adequacy of the invariant set itself.

Key Factorizations

Separation	Failure mode made visible
World state w vs. admitted representation $\hat{x}$	Context misrepresentation, poisoned or missing evidence
Proposal $\hat{u}$ vs. executable action a	Proposal laundering, runtime drift, tool substitution
Operational gate $\hat{G}$ vs. independent criterion G^*	Evaluator gaming, self-grading, false admission / rejection
Local checks vs. trajectory invariants Ψ	Cumulative scope creep, multi-step authority or resource violations
Compliance relative to model M vs. invariant adequacy	Wrong, incomplete, stale, or illegitimate invariant sets

Adaptive Escape as a Hazard Process

A fixed false-admission rate is not a stable description of an adaptive proposer. Define the conditional escape hazard on the t -th invalid gate-facing query (conditional on no earlier escape):

$$h_t = P(E_t \mid \text{no earlier escape}) \rightarrow q_M = 1 - \prod_{t=1..M} (1 - h_t)$$

The constant-hazard baseline $h_t = \varepsilon$ recovers $q = 1 - (1 - \varepsilon)^M$. The empirical question is whether h rises as an adaptive proposer learns the gate's weak regions. Average pass rates alone may fail to characterize risk under search pressure.

Observability-Aware Error Decomposition

Within a preregistered scenario class c , false-admission error is partitioned by whether decisive evidence was present in the admitted representation ($O = 1$) or absent ($O = 0$):

$$\varepsilon_{FA}^{(c)} = P(O=0 \mid G^*=0, c) \cdot \varepsilon_{miss}^{(c)} + P(O=1 \mid G^*=0, c) \cdot \varepsilon_{judge}^{(c)}$$

This separates context-admission failures from judgment failures. Mixture weights are properties of the benchmark or deployment distribution, not intrinsic constants of the gate.

Primary Measurement Targets

- Independent invalid-proposal rate and independently labeled executed violations (held-out G^*).
- Conditional escape hazard h_t and episode escape probability q_M under rising search budget M and proposer class κ .
- False-admission / false-rejection rates partitioned by observability within scenario class.
- Runtime fidelity (canonical permit vs. executed action) and trajectory-level invariant violations.
- Capability, rare-case competence, truthful dissent, and human intervention burden measured jointly — lower human load counts as maturation only when independent violation rate does not rise.

Empirical Stance & Boundary Conditions

Experiments should compare unconstrained optimization, fixed binary-violation penalties, constrained/shield baselines, and ICTF-style external gating (with and without controlled internalization) across an explicit capability and search schedule. Internalization success is scored only against independent labels, never against the operational gate alone.

- ICTF claims no additional expressive power beyond sufficiently augmented state models.
- Model-relative compliance is not alignment: a system can satisfy every specified invariant and still be misaligned if the invariant set is incomplete, illegitimate, or stale. The wrong-invariant problem lies outside the formal guarantee.
- Mathematical content is deliberately modest (standard survival identities, hazard decomposition, observability partitioning, forward-invariance under gate error). Distinctive claims are empirical and architectural.